



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP2523: DATABASE

SEM I 2024/2025

DATABASE DESIGN DESCRIPTIONS : CONCEPTUAL DATABASE DESIGN

FOR <KADA Cooperative System>

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1.0 Introduction

Kemubu Agricultural Development Authority is a government agency under the aegis of the Ministry for Agriculture and Food Security to alleviate the socio-economic status of farmers within the Kemubu region, Malaysia. Established to support agricultural development and community welfare, KADA has been a driver in modernizing farming and making financing available through cooperative membership and loan programs.

Despite its importance, KADA's system on cooperative membership and loan applications currently operates on a purely manual and paper-based process, which is increasingly inefficient, data loss, and environmentally unsustainable. Thus, this has resulted in slow processing of applications, mismanagement of data, or reliance on practices no longer applicable in the digital world.

In this regard, KADA has developed an innovative, web-based platform called the "KADA Cooperative System." This system will digitize membership and loan applications, automate processes of verifying eligibility, improve data security, enhance administrative efficiency, and contribute toward environmental sustainability. The system is user-friendly and will be accessible to KADA staff and members of the cooperative only, ensuring the security of sensitive financial processes.

Through digital transformation, KADA is looking forward to continuing its mission of empowering farmers with innovative, efficient, and sustainable solutions that meet modern agricultural and socio-economic demands.

1.1 Overview about the company

The Kemubu Agricultural Development Authority, a statutory body under the Ministry of Agriculture and Food Security, was established to develop the socio-economic well-being of farmers in the region of Kemubu, Malaysia. KADA plays a pivotal role in supporting agricultural development, improving farming practices, and providing financial and cooperative assistance to its members.

With the aim of promoting the livelihoods of the farmers, KADA focuses its endeavors on modernization in farming techniques, access to facilities, and community welfare. In support, the agency has its cooperative arm that addresses loan schemes and membership designed for financial stability and encourages farmers to practice sustainable agriculture.

KADA, through its vision, has continued to address the challenges in its operations, part of innovation and efficiency. It also seeks to empower farmers with technological advancements and environmentally conscious practices for the long-term sustainability of agriculture and economic activities within the region.

1.2 Problems

The existing manual system of cooperative operations has many limitations which affect its efficiency and accessibility. Below, I address some of the problems faced by the system.

1. No Online Loan Tracking

Loan applicants do not have the facility to know the status of their applications online. They visit the office or contact members of staff directly. All these are leading to losses in time and delays.

2. Limited Dividend Information Access

There is no real-time information access concerning dividend calculations and eligibility. Members have to create follow-ups for themselves, which is very inconvenient and requires much time.

3. Dependency on Physical Record Storage

Physical storage of documents risks loss, damage, or misplacement. This method also hinders data access and retrieval, making the process sluggish and inefficient.

4. Inaccessibility of Savings and Reports

Savings and annual reports are not available online, meaning members have to go to the cooperative office to access them. This leads to delays, especially for those far from the office.

5. No Mechanism to Detect Fraud

The manual system lacks an automated framework to detect or prevent fraudulent activities in regard to false declarations of income. This makes the verification and validation of information vulnerable during the process.

These challenges show the inefficiency of the existing manual system and highlight the need for a digital solution to improve accessibility, transparency, and operational effectiveness while minimizing risks.

1.3 Proposal

The Kemubu Agricultural Development Authority (KADA) is currently operating its cooperative membership and loan application process manually. This system is inefficient, prone to data loss, and environmentally unsustainable. In light of these challenges, it is proposed to implement a comprehensive, all-digital platform to streamline operations, enhance security, and improve accessibility for KADA's staff and cooperative members.

1. Creation of an All-Digital Integrated Platform

- **Online Application and Registration:** A user-friendly website will be developed, allowing KADA employees to electronically apply for membership loans and register for cooperative membership. Both procedures will be combined in one smooth workflow on the platform.
- **Verification of Eligibility:** An automated system will be put in place for the verification of the eligibility of cooperative members. It would cross-match employee data against predefined criteria for speedier and more accurate eligibility determinations.
- **Electronic Submission and Monitoring of Forms:** The platform shall allow employees to submit forms digitally from anywhere at any time. A tracking system shall be in place to show the real status of applications in real time.

2. Effective and Secure Data Management

- **Centralized Database:** Create a centralized digital database to securely store all data about the members and loans, hence limiting data loss, damage, or misplacement to almost nil.
- **Data Encryption and Privacy:** Robust security, including data encryption, should be ensured so that the personal and financial information of KADA employees is safe, maintaining privacy and thereby building trust in the new system.

3. Improved Convenience of Access

- **User Identification and Secure Access:**
It will be designed with login to ensure only authorized personnel can access and modify sensitive information, adding to the security of the system.

4. Enhanced Efficiency in Administration

- **Automated Board Member Workflow:** The website will reduce the workload for the cooperative board members by providing easy access to online forms and applications. The system will also include features to help in organizing, filtering, and reviewing applications with efficiency during the monthly meetings.

5. Environmental and Financial Benefits

- Paper Use Reduction: A fully digital platform will reduce paper use tremendously, hence promoting environmental sustainability and reducing the operational costs which are related to paper and storage.
- Storage Savings: There would not be any need for any physical storage of records on paper, freeing up invaluable space and reducing the related costs of storing and management of paper records.

The inefficiency and limitation of the present manual system will be addressed by this digital platform. The proposed solutions will improve data security, enhance operational efficiency, and promote environmental sustainability. With this advanced digital system, KADA will be able to provide more convenience for its members and further improve overall organizational performance in supporting the continued success of the agency's mission.

1.4 Definitions, Acronyms and Abbreviations

Table 1.1 : Definitions, Acronyms and Abbreviation

Terms	Definitions
KADA	Kemubu Agricultural Development Authority
BOD	Board of Director
SD	System Documentation
Admin	Administrator

2.0 Database System Requirement

2.1 System Overview

The KADA Cooperative System is a new web-based platform that is integrated from the old system which is designed for reducing the manual work such as paper-based records by the cooperative. The main functions of the system include the handling of the membership details, loan applications, guarantor details, and financial statement to help in the management of the cooperative. The system is accessible by the staff, admin, and BOD of KADA Cooperative where the staff can apply for membership and loan application, the admin able to track and manage their applications, while the BOD able to review and approve all the applications. Furthermore, our system also considered data security and integrity by adding an authentication function to prevent any unauthorized users from entering the system. Overall, the system aims to streamline the cooperative operations and enhance transparency for all the users to use the system.

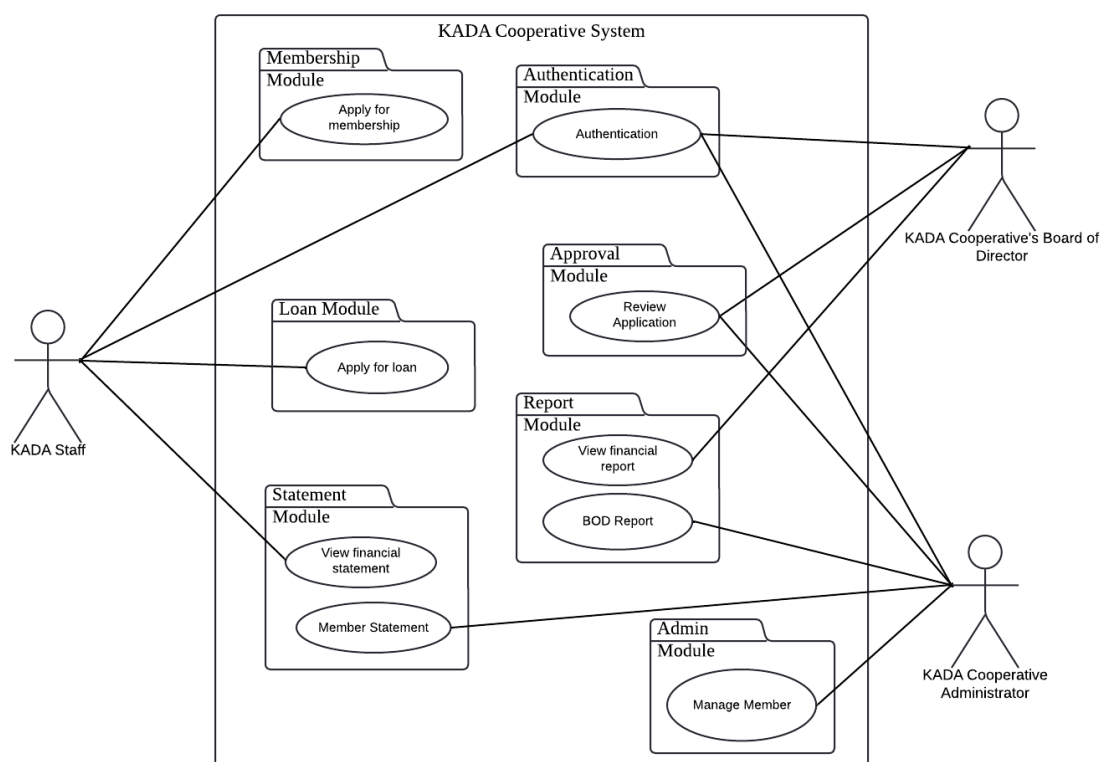


Figure 2.1: Use Case Diagram for KADA Cooperative System

2.2 Module 1 : Authentication

Nabil Aflah Boo is responsible for this module.

2.2.1 Module Description

This module involves the staff, admin and BOD to login, reset password, and register membership to access the system interface according to their roles. This module is important as they need to be authorized to be allowed for using the overall system.

2.2.2 Transaction Requirement: Data Entry

1. Users enter their email address and password for the authentication.
2. Users reset their password by entering their email address and the new password for the system.
3. The non-member of the KADA need to register for the membership (wait to be approved by BOD).

2.2.3 Transaction Requirement: Data Update/Deletion

1. Update/delete the email address and password for the authentication.
2. Update/delete the email address and the new password.
3. Update/delete the details of membership registration.
4. Update/delete the unapproved membership.

2.2.4 Transaction Requirement: Data Queries

1. List of the users according to their name, id, email, password, role, and registration date.
2. List of the unapproved membership application.
3. List of the applicant's name and their details (haven't been approved by BOD).

2.2.5 Local Conceptual Data Model (ERD) for Module 1

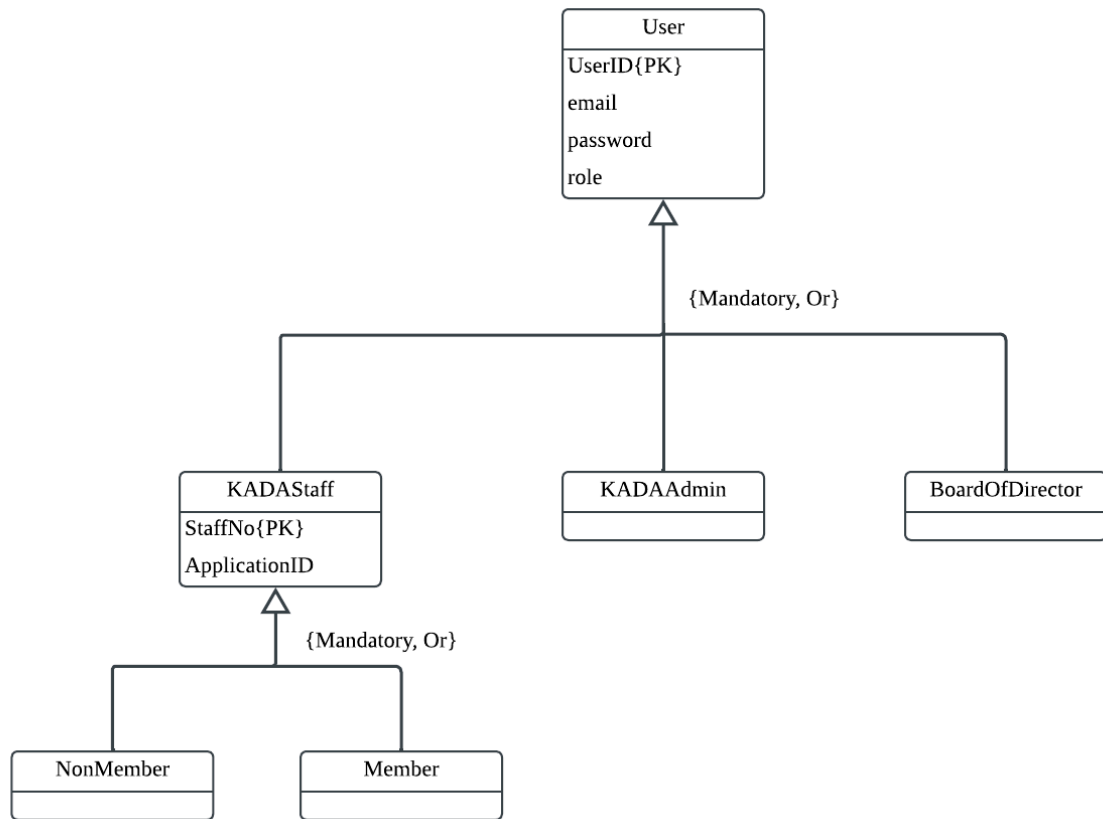


Figure 2.2: Local ERD for Module 1

2.3 Module 2 : Membership

Nabil Aflah Boo is responsible for this module.

2.3.1 Module Description

This module deals with registering new members into the KADA Cooperative Systems. All of data entering, updating, removal, and inquiries about the membership information is handled by this module. The applicants are allowed to enter their details online and the membership is granted once approved.

2.3.2 Transaction Requirement: Data Entry

1. Applicants enter details: Name, IC number, contact details, address, and employment details.
2. Validate and securely save the data in the database.

2.3.3 Transaction Requirement: Data Update/Deletion

- Update
 1. Modify member information (e.g. : phone number, address, employment status).
 2. Making sure changes are logged every time for auditing reasons.
- Deletion
 3. Erase membership information for inactive members based on predefined criteria (e.g. : resignation, inactivity).
 4. This action should be verified to avoid any accidental deletion.

2.3.4 Transaction Requirement: Data Queries

1. Obtain member information through membership ID or name.
2. Search a member by different filters, such as address or employment status.
3. Generate summary reports about all approved members for administrative requirements.

2.3.5 Local Conceptual Data Model (ERD) for Module 2

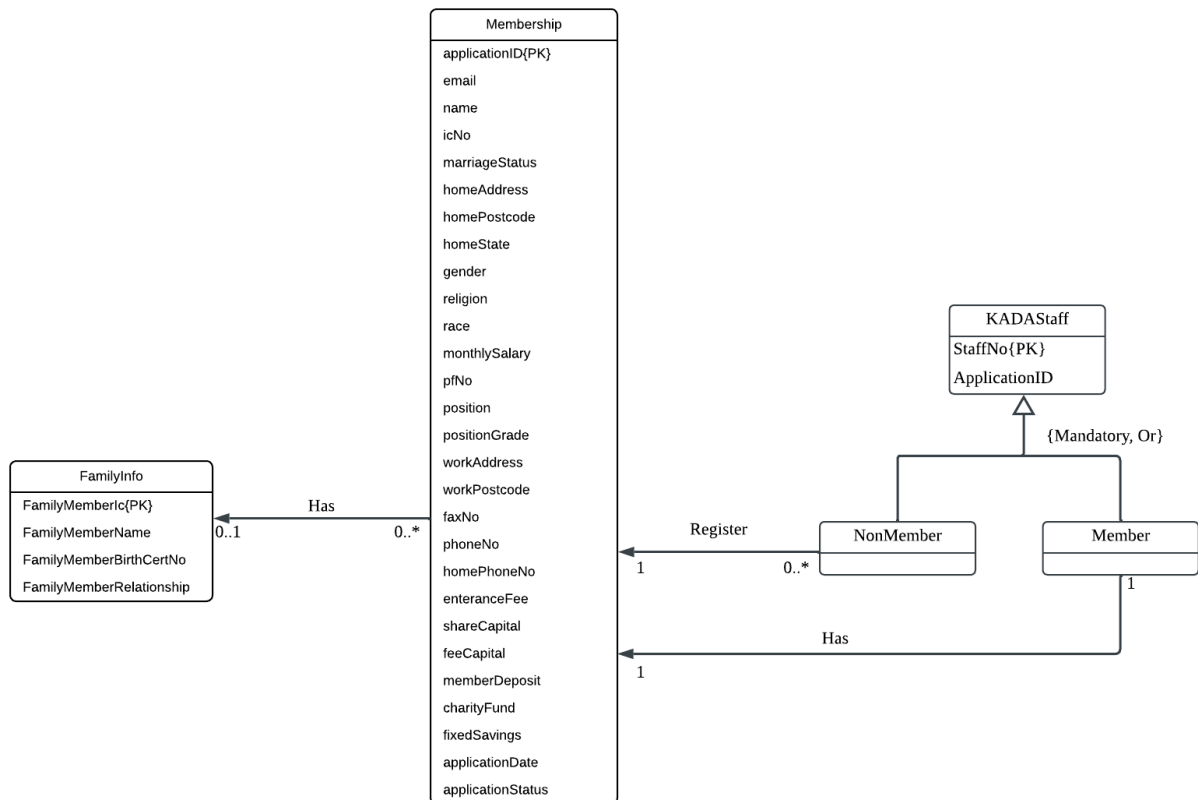


Figure 2.3: Local ERD for Module Membership

2.4 Module 3 : Loan

Lubna Al Haani is responsible for this module.

2.4.1 Module Description

This module handles the process of applying for loan applications which contain a use case named “Apply for loan”. It ensures a streamlined process for members to apply for loans and provides resources needed for members to view the status of their applications.

2.4.2 Transaction Requirement: Data Entry

1. Member selects the loan type and input details (loan amount, duration, guarantor details, etc..)
2. System validates input and submits the application.

2.4.3 Transaction Requirement: Data Update/Deletion

1. System update loan application status after application is reviewed by board of directors.

2.4.4 Transaction Requirement: Data Queries

1. List the list of loan applications with its status.
2. Display all the application details.

2.4.5 Local Conceptual Data Model (ERD) for Module 4

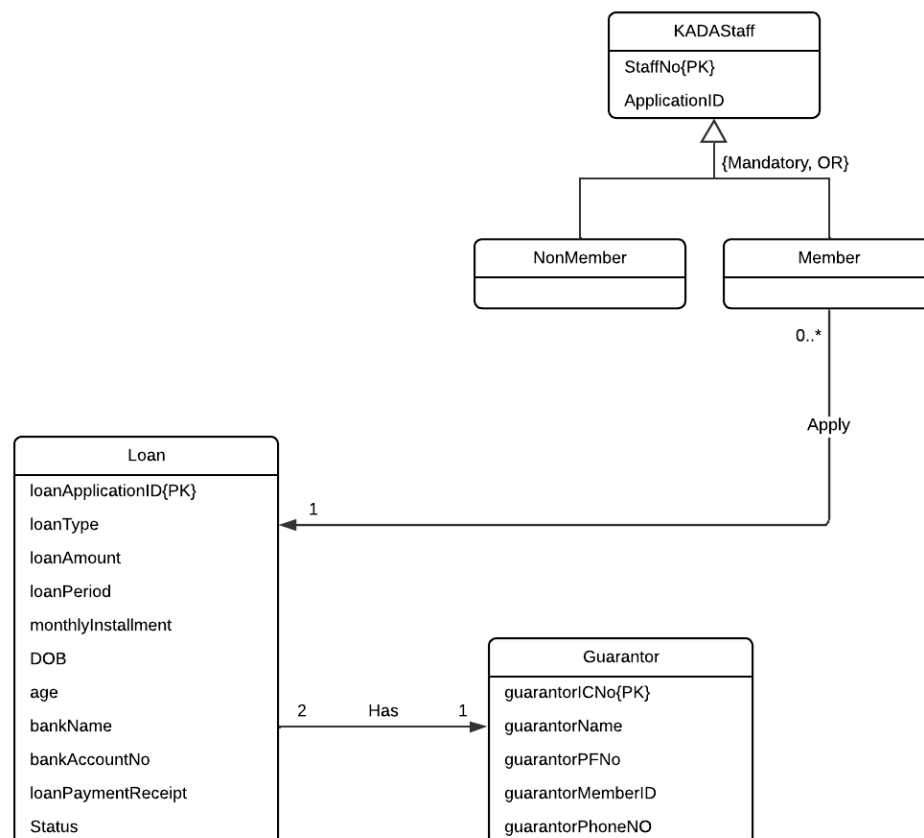


Figure 2.4: Local ERD for Module 4

2.5 Module 4 : Statement

Pravinraj A/L Sivabathi is responsible for this module.

2.5.1 Module Description

The Member Statement module allows users to safely get access to their statements according to their credential log-in and selects. This module ensures the member can access the statement needed savings balance or loan balance, among others. The system accesses it and displays the latest data, which is available, right in real time to the member. Plus, the module also allows members to save the statement in any format they want, even PDF; therefore, convenience and flexibility are left to the user's decisions while managing their financial information.

2.5.2 Transaction Requirement: Data Entry

1. Members can select the type of statement which is a savings balance or statements of loan balance to view.

2.5.3 Transaction Requirement: Data Update/Deletion

1. System update new member statement monthly to keep and save new member statement's records.

2.5.4 Transaction Requirement: Data Queries

1. Retrieve and display specific member statements, such as savings or loan balances.
2. Provide options to export the statement in PDF or other format for personal records or further use.

2.5.5 Local Conceptual Data Model (ERD) for Module 5

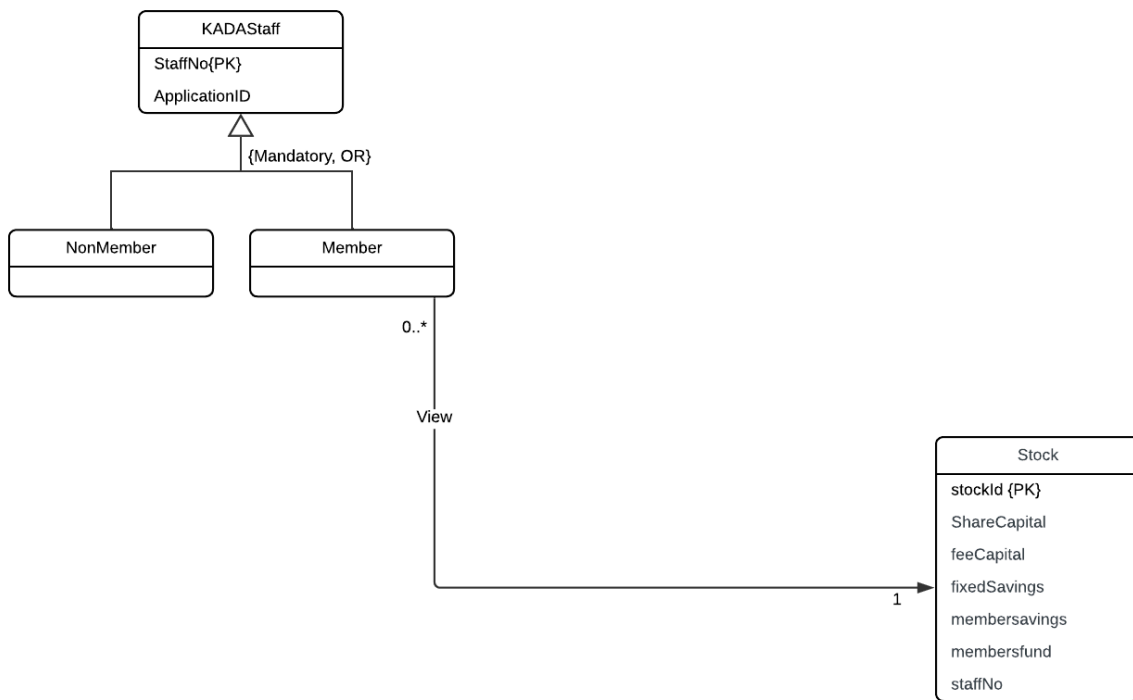


Figure 2.5: Local ERD for Module 5

2.6 Module 5 : Report

Nurul Adriana is responsible for this module.

2.6.1 Module Description

This module handles the report compilation for the Board of Directors to aid them in decision-making. It combines both financial and operational data into well-arranged reports, including membership numbers, loan statuses, and the overall performance of the cooperative.

2.6.2 Transaction Requirement: Data Entry

- Administrator Input:
 1. Enter financial data (e.g. : monthly income, savings, and loan payments).
 2. Update transactions related to members (e.g. : loan disbursement information).

2.6.3 Transaction Requirement: Data Update/Deletion

- Update
 1. Modify report settings (e.g. : time-frame, type of analysis).
 2. Make corrections to any wrongly entered data (e.g. : financial errors).
- Deletion
 1. Delete outdated reports (e.g. : monthly or quarterly data) according to archive requirements.

2.6.4 Transaction Requirement: Data Queries

1. Generate reports that detailed on;
 - Total of active memberships.
 - Loan status; approved, pending, denied.
 - Financial report (e.g. : profit, loss statements).
2. Perform query data patterns over a given period of time (e.g. : membership growth, the number of loan application trends).
3. Provide ability to generate and export reports in PDF or Excel format for presentations and analysis.

2.6.5 Local Conceptual Data Model (ERD) for Module 5

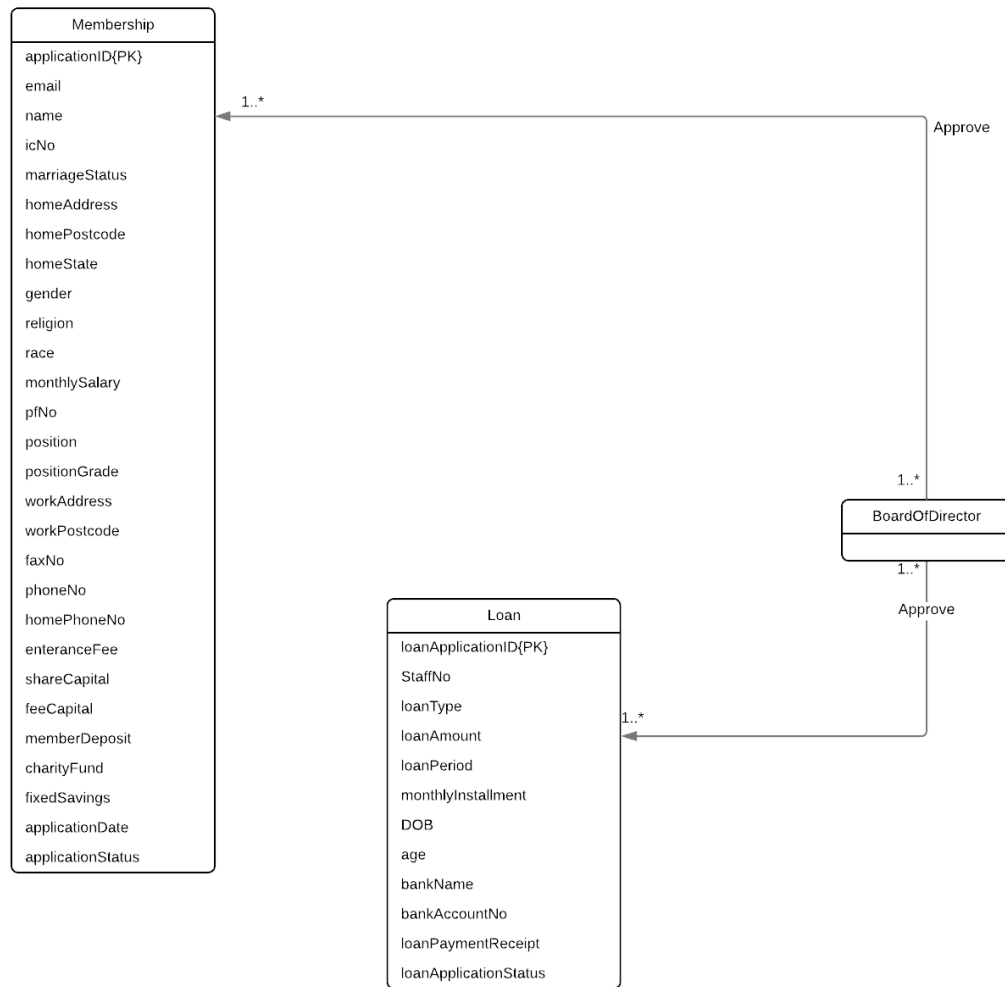


Figure 2.6: Local ERD for Module 6

2.7 Module 6 : Admin

Nur Firzana is responsible for this module.

2.7.1 Module Description

This module enables KADA admins to manage member information including the KADA cooperative profiles, dashboard and member statement, oversee the application statuses and update the relevant reports. It supports accurate record-keeping and reliable data management. It also plays a central role in supporting the Report Module and Approval Module by ensuring data accuracy and integrity.

2.7.2 Transaction Requirement: Data Entry

1. Enter applicants' names to easily oversee their information details.
2. Enter year and month to edit the member statement.

2.7.3 Transaction Requirement: Data Update/Deletion

1. Update/Delete the public KADA cooperative profile, dashboard and member statements.
2. Update/Delete application status including the membership and loan application that has been discussed with BOD.
3. Update/Delete overall report that synchronized with system's latest data to ensure the BOD can review it easily.

2.7.4 Transaction Requirement: Data Queries

1. List the membership application with their information.
2. List the loan application with their information.
3. List the member statement according to the membership name.
4. List the application status (approved/disapproved) along with the applicants' details.
5. List the overall report based on the yearly report.

2.7.5 Local Conceptual Data Model (ERD) for Module 6

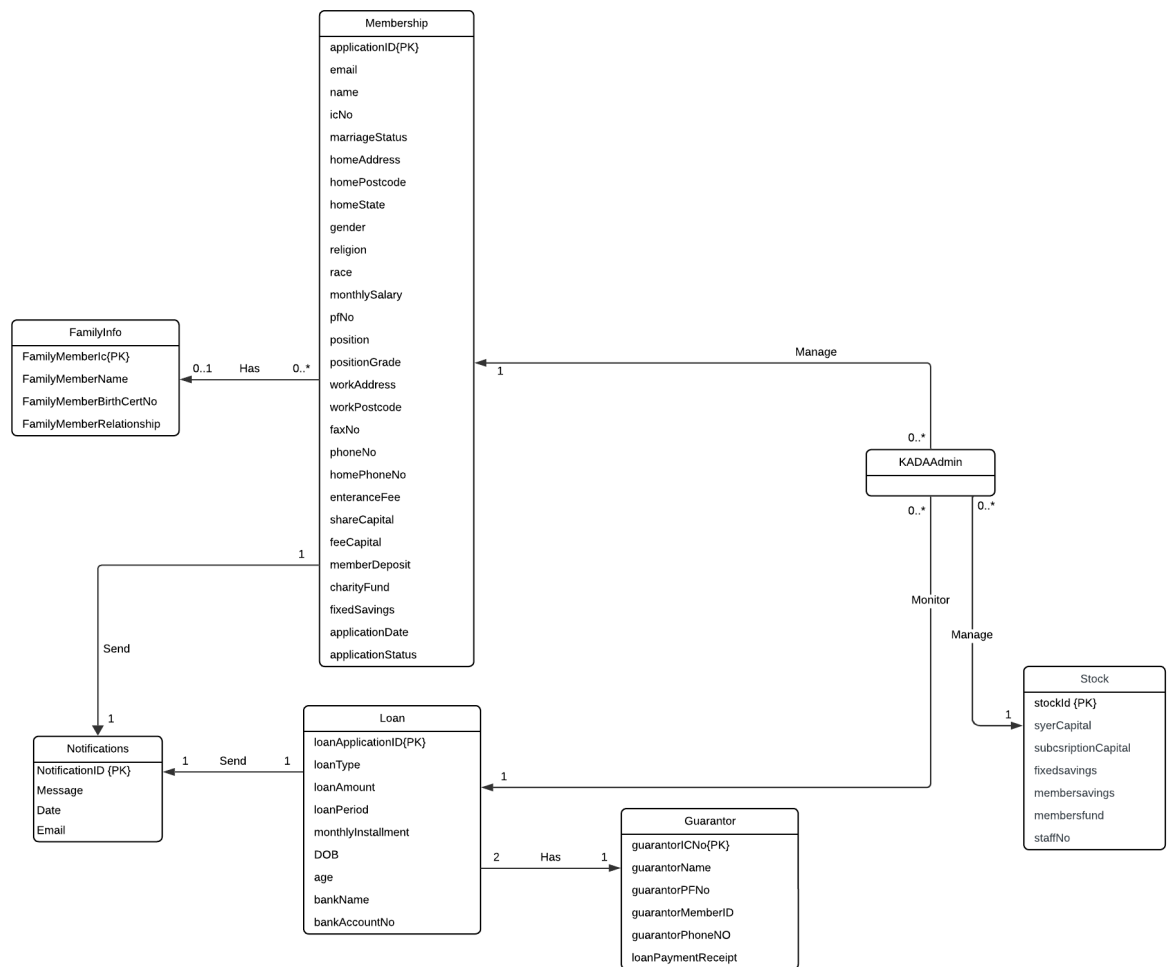


Figure 2.7: Local ERD of Module 6

2.8 Module 7 : Approval

Lubna Al Haani is responsible for this module.

2.8.1 Module Description

This module manages the process of reviewing and approving/rejecting member and loan applications. It provides the tools for the board of directors to view and approve/reject applications. Admin also provided tools to monitor the applications status. The use case involved is "Review Application".

2.8.2 Transaction Requirement: Data Entry

1. Boards of directors approve/reject for each application with comment if needed.

2.8.3 Transaction Requirement: Data Update/Deletion

1. System update application status.

2.8.4 Transaction Requirement: Data Queries

1. List all applications according to application type.
2. Display all application details.

2.8.5 Local Conceptual Data Model (ERD) for Module 7

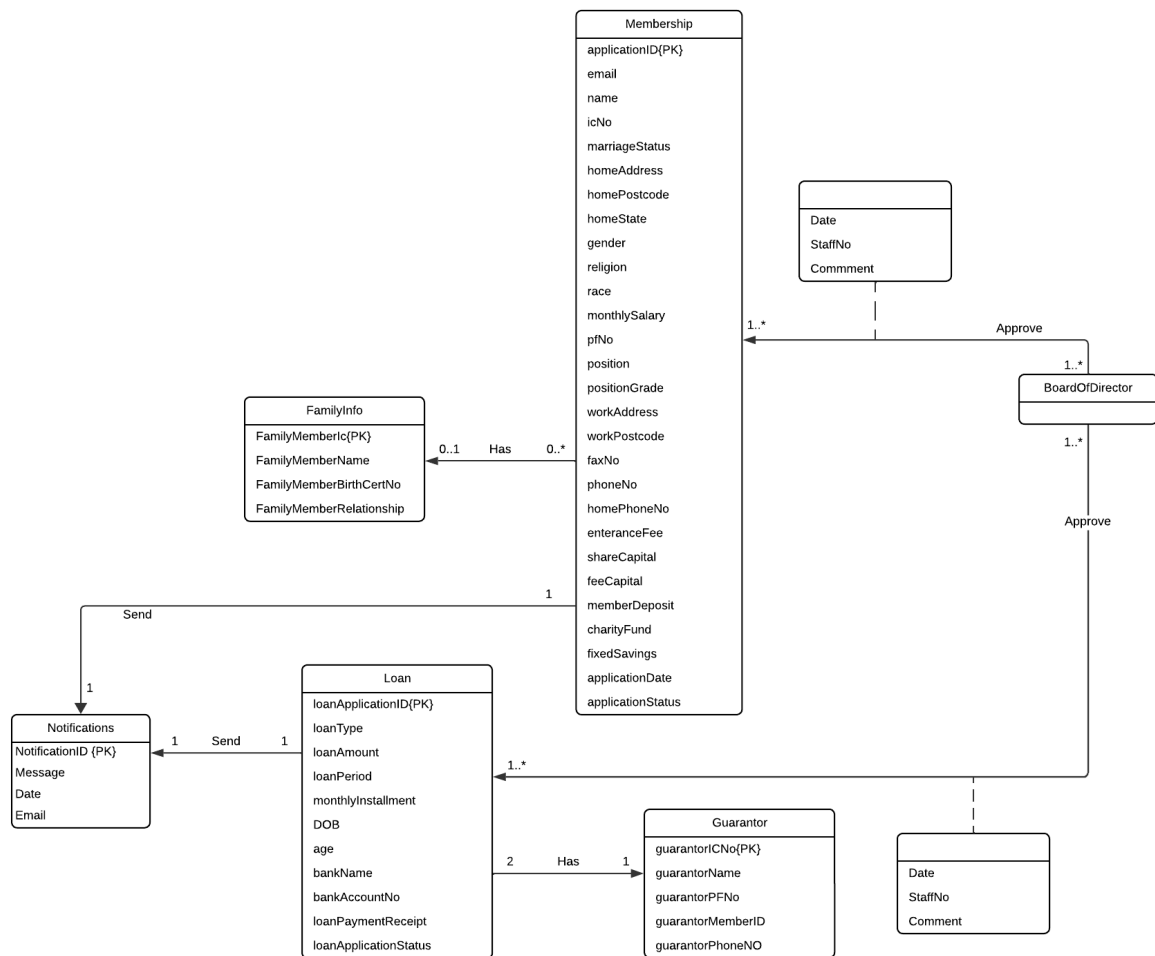


Figure 2.8: Local ERD for Module 7

3.0 Global Conceptual Data Model (Global ERD)

The global conceptual ERD is created by combining the local conceptual models from the 7 modules of the system, as described in Section 2.0. This global ERD consists of 12 entity types and 17 relationship types. Each entity and attribute is crucial for storing the system's data. In this section, we will enhance the ERD by applying advanced techniques like generalization and specialization. Additionally, the transaction path will be employed to identify the unnecessary relationships, redundancies and missing connections that could stop the system functionality .

3.1 Global Conceptual Design (ERD)

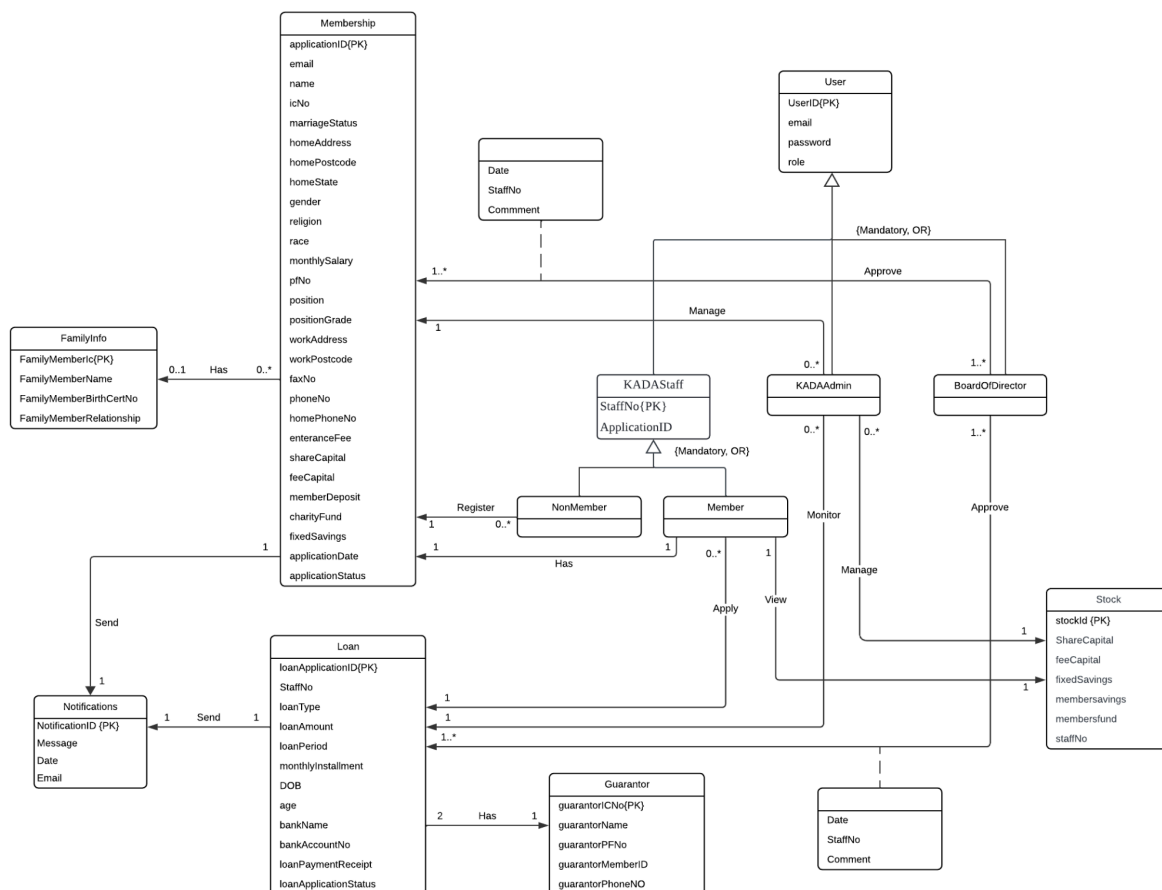


Figure 3.1 : Global ERD of KADA Cooperative System

3.2 Verified Global Conceptual Design (Verified ERD)

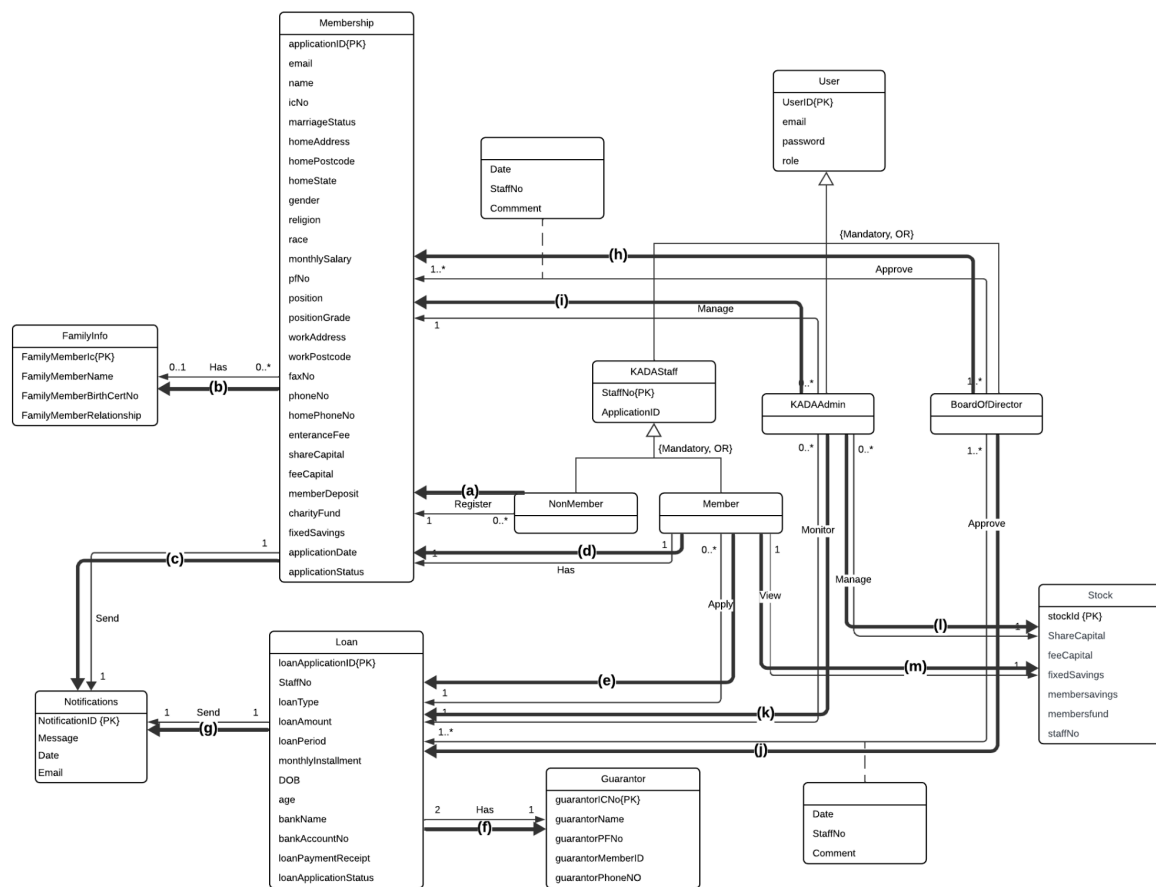


Figure 3.2: Verified ERD

Symbol	Transaction
a	2.3.2.1, 2.3.2.2
b	2.3.2.1
c	2.3.2.2
d	2.3.3.1, 2.3.3.2, 2.3.3.3, 2.3.3.4
e	2.4.2.1, 2.4.2.2, 2.4.4.1, 2.4.4.2
f	2.4.2.1
g	2.4.3.1
h	2.8.2.1, 2.8.3.1, 2.8.4.1, 2.8.4.2
i	2.7.2.1, 2.7.3.1, 2.7.3.2, 2.7.4.1, 2.7.4.4

j	2.8.2.1, 2.8.3.1, 2.8.4.1, 2.8.4.2
k	2.7.2.1, 2.7.3.2, 2.7.4.2, 2.7.4.4
l	2.7.2.1, 2.7.2.2, 2.7.3.1, 2.7.4.3
m	2.5.2.1, 2.5.3.1, 2.5.4.1, 2.5.4.2

3.3 Data Dictionary for the global conceptual ERD

Attribute	Description	Data Type	Length/Size	Constraints
Member_ID	Unique identifier for each cooperative member	INT	10	Primary Key, NOT NULL
Name	Full name of member	VARCHAR	50	NOT NULL
IC_Number	Member's identification card number	VARCHAR	12	UNIQUE, NOT NULL
Contact_Details	Phone number or email of the member	VARCHAR	100	NOT NULL
Address	Residential address of the member	TEXT	-	-
Employment_Details	Job-related information for the member	VARCHAR	255	-
Loan_ID	Unique identifier for each loan application	INT	10	Primary Key, NOT NULL
Loan_Amount	Amount requested for the loan	DECIMAL	10,2	NOT NULL
Loan_Type	Type of the loan (e.g., personal, business)	VARCHAR	50	NOT NULL

Application_Status	Status of the loan application	VARCHAR	20	NOT NULL, DEFAULT, 'Pending'
Guarantor_Details	Information about the loan guarantor	VARCHAR	50	NOT NULL
Statement_Type	Type of financial statement (e.g., savings)	VARCHAR	50	NOT NULL
Statement_Date	Date the statement was generated	DATE	-	NOT NULL
Report_ID	Unique identifier for reports	INT	10	Primary Key, NOT NULL
Report_Type	Type of the report (e.g., financial, loan)	VARCHAR	50	NOT NULL
Admin_ID	Unique identifier for administrators	INT	10	Primary Key, NOT NULL
Approval_Status	Approval/rejection status by the board	VARCHAR	20	NOT NULL
Approval_Comments	Comments provided during approval	TEXT	-	-

4.0 Appendices

In this section, it provides the use case diagram for the whole system, activity diagram for every use case identified which is the 7 modules and meeting log.

4.1 Use case Diagram for the system

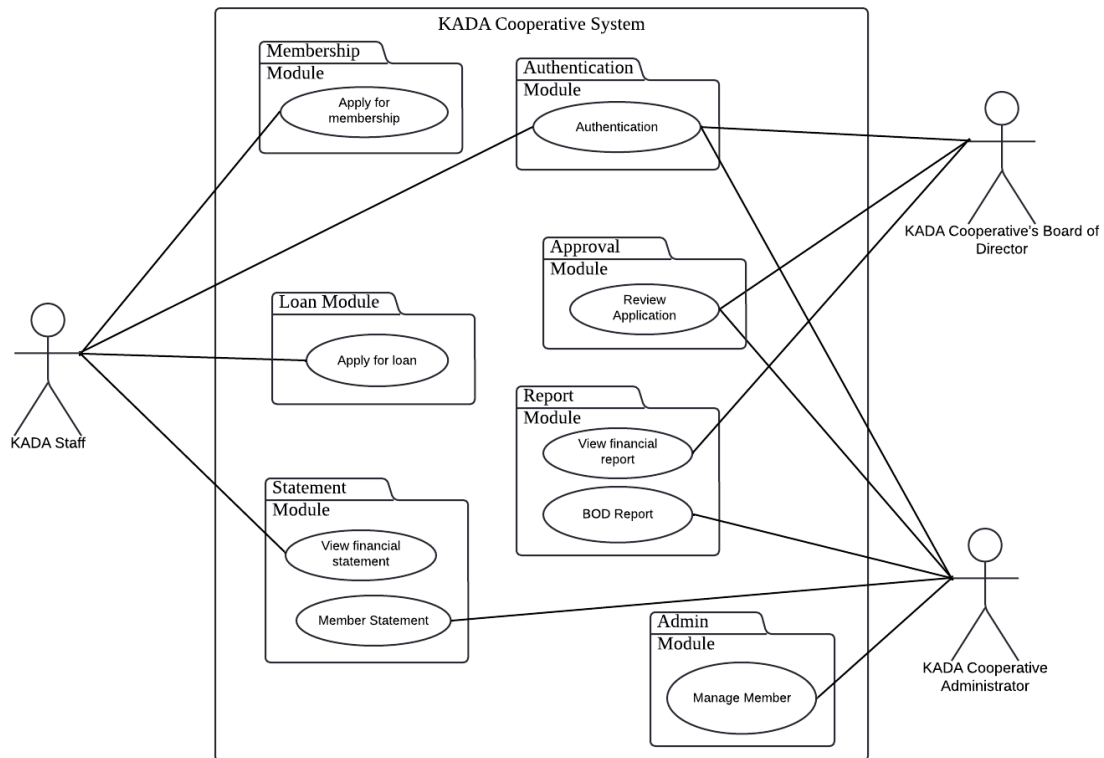


Figure 4.1: Use Case Diagram for KADA Cooperative System

4.2 Activity Diagrams for every use case identified

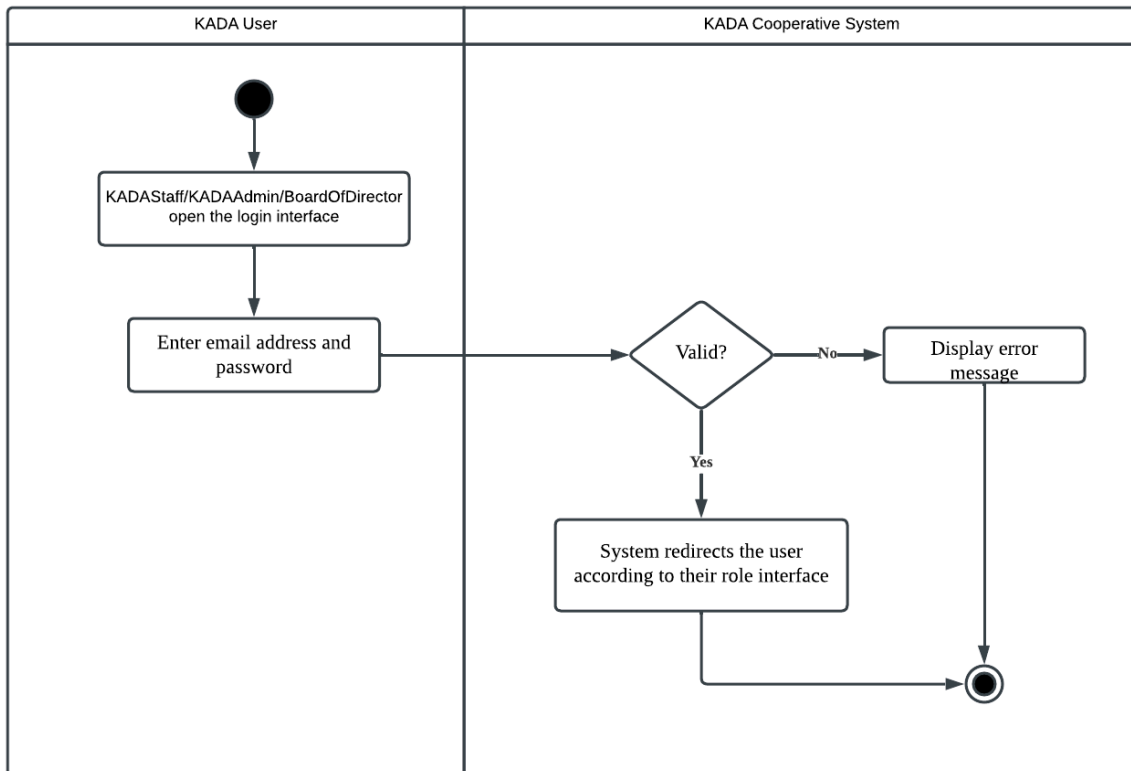


Figure 4.2.1 : Activity Diagram for Module Authentication

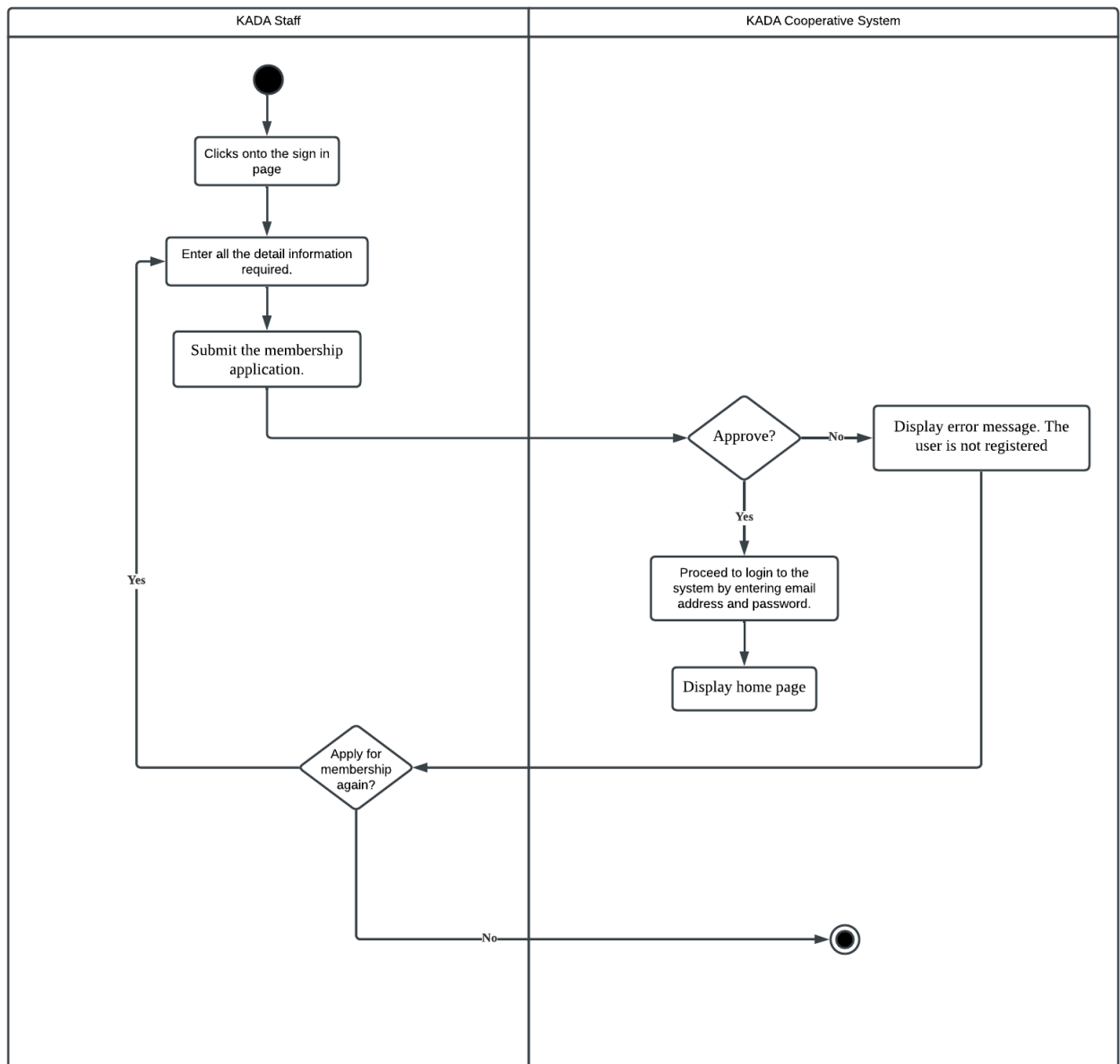


Figure 4.2.2 : Activity Diagram for Module Membership

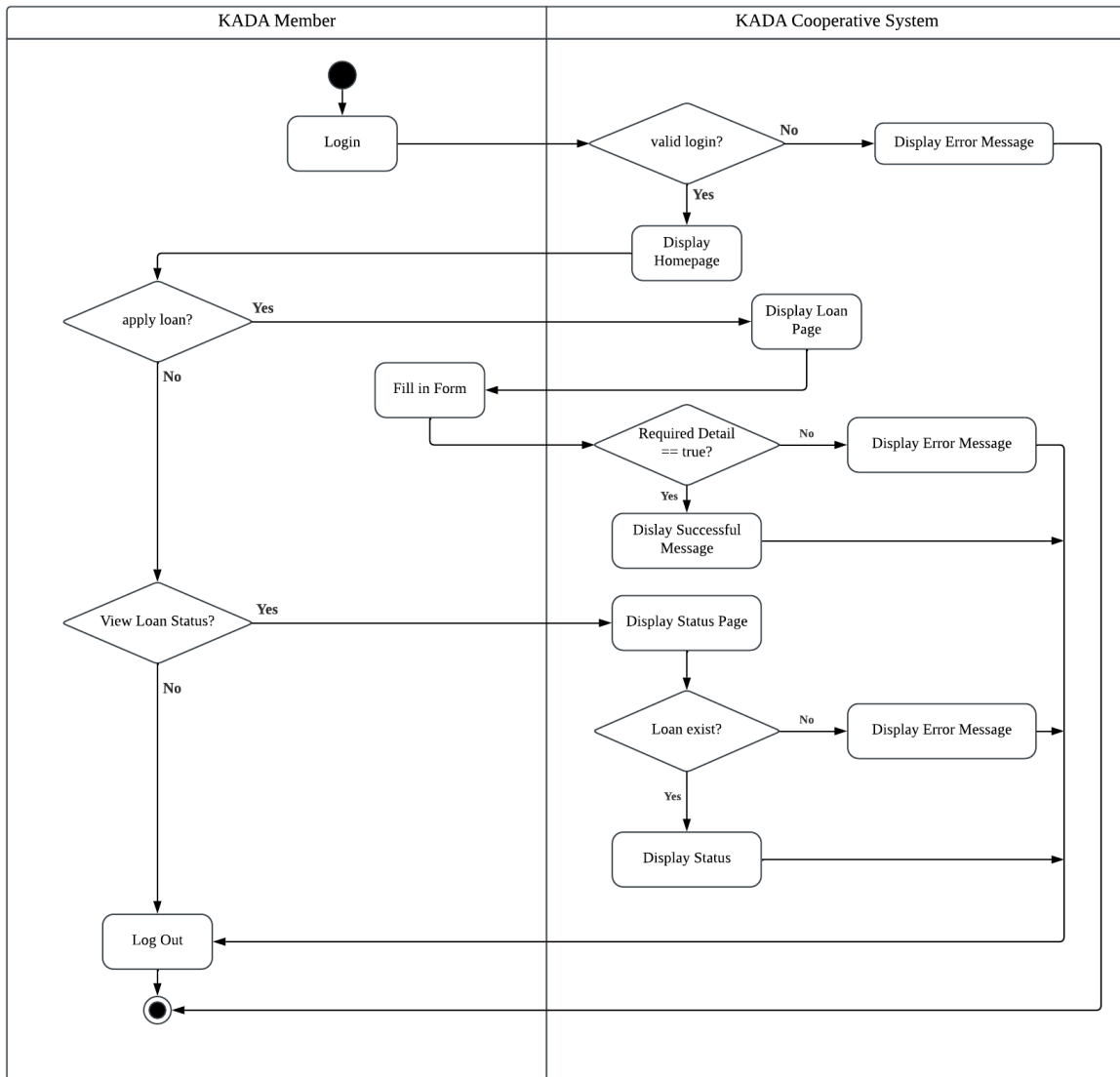


Figure 4.2.3 : Activity Diagram for Module Loan

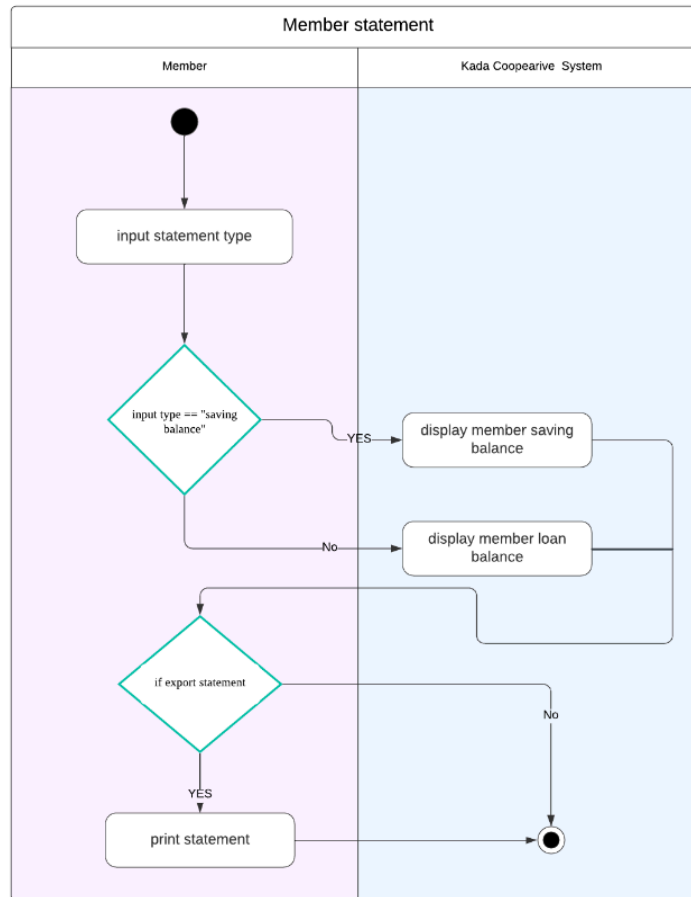


Figure 4.2.4 : Activity Diagram for Module Statement

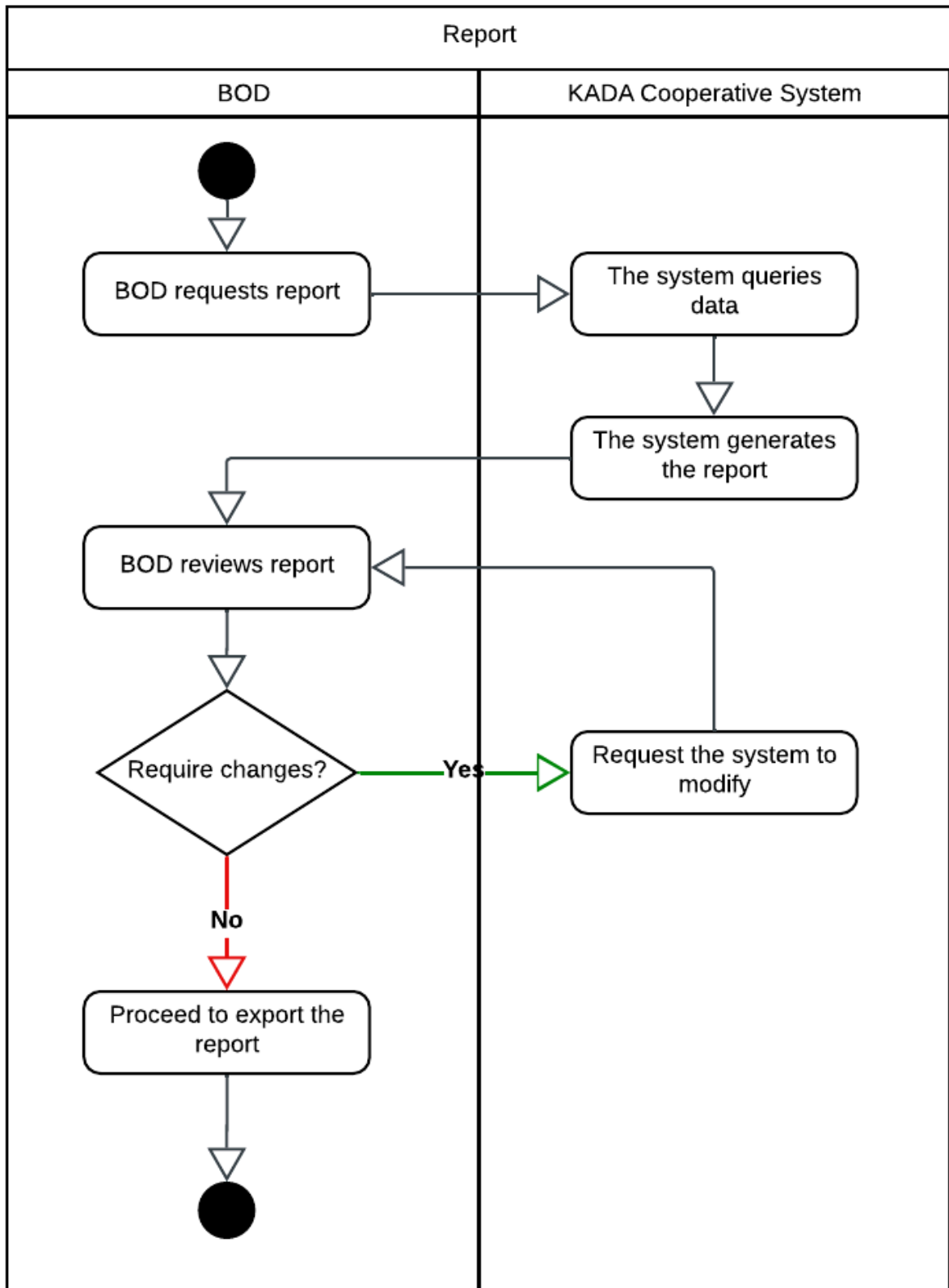


Figure 4.2.5 : Activity Diagram for Report

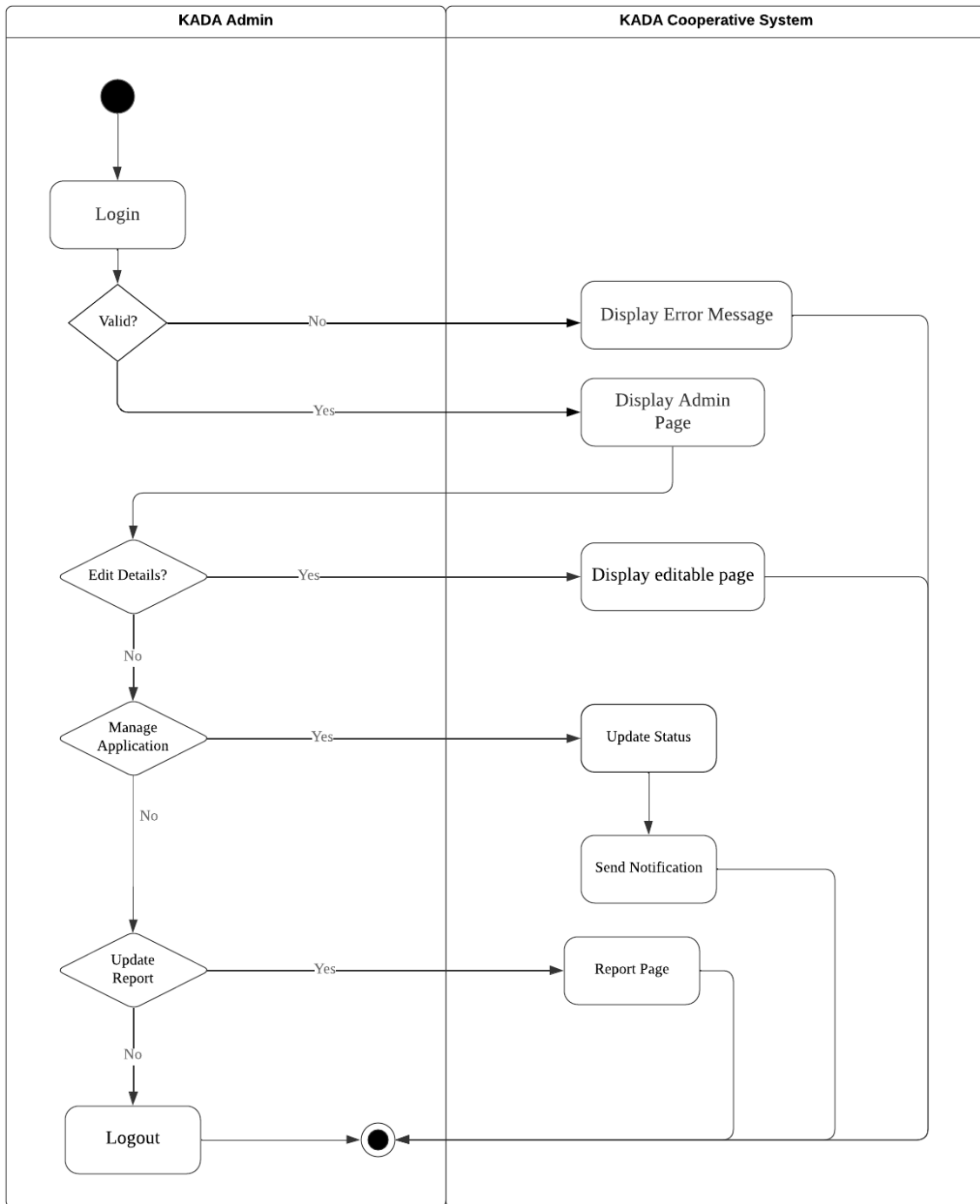
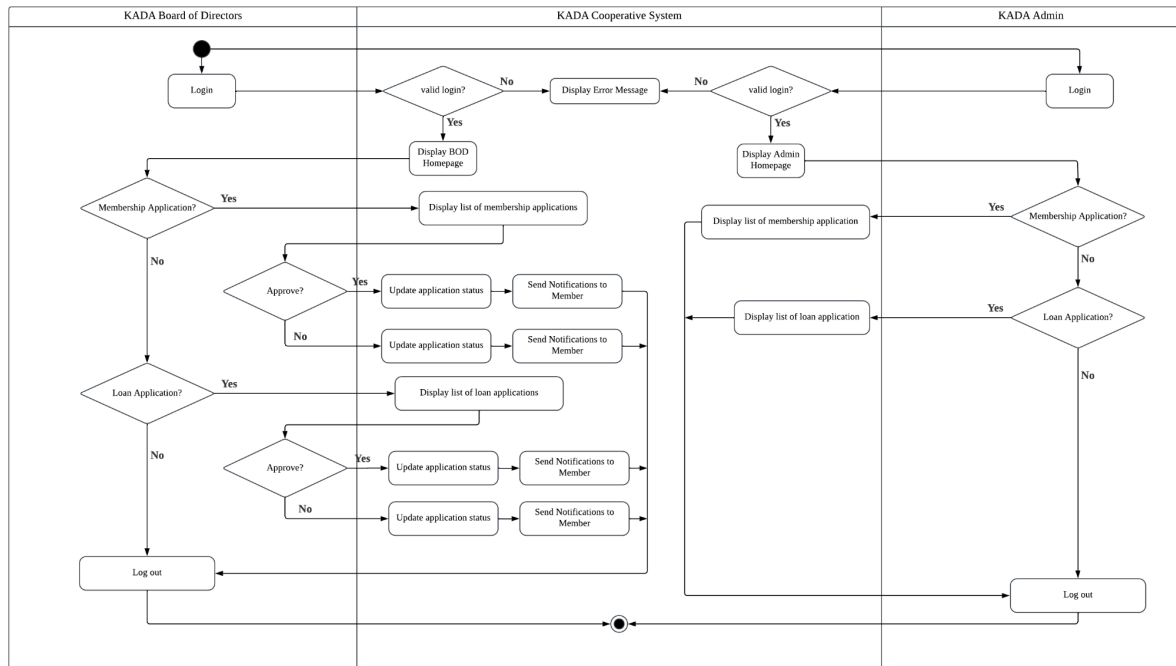


Figure 4.2.6 : Activity Diagram for Module Admin



4.3 Meeting Log

Meeting Title: P2 Documents Discussion.

Date: 16/12/2024

Time: 9.50 p.m

Location: Virtual Meeting

Attendees:

- Lubna Al Haani binti Radzuan
- Nur Firzana Binti Badrus Hisham
- Nabil Aflah Boo Binti Mohd Yosuf Boo Yong Chong
- Nurul Adriana Binti Kamal Jefri
- Pravinraj A/L Sivabathi

Meeting Objectives:

The objective of this meeting is to divide tasks between team members and discuss the content of P2 documents that need to be done.

Meeting Agenda:

1. Task distribution among the team members.
2. Discuss the contents of the documents.

Discussion Summary:

1. Task is distributed to the team members according to their preferences.
2. Some of the content needs to be confirmed by the client to ensure the correct information is used to develop the design for the system.